

Homework – SMPC Mini-Project 2

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1 & 2: LC Filter Cut-off Frequency

- **Nominal Cut-off Frequency (f_c):** Given $L = 10\mu\text{H}$ and $C = 1.0\mu\text{F}$:

$$f_c = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10\mu\text{H} \cdot 1\mu\text{F}}} \approx 50.33 \text{ kHz}$$

- **Tolerance Analysis (20%):**

- **Minimum f_c :** $L = 12\mu\text{H}$, $C = 1.2\mu\text{F} \implies f_{\min} \approx 41.94 \text{ kHz}$
- **Maximum f_c :** $L = 8\mu\text{H}$, $C = 0.8\mu\text{F} \implies f_{\max} \approx 62.91 \text{ kHz}$

- **Resulting Range:** 41.94 kHz to 62.91 kHz.

3: Voltage Control Loop Design

- **Specifications:**

- **Loop Gain at 200 Hz:** Target > 40 dB.
- **Stability:** PM $> 60^\circ$.
- **Passive Constraints:** All $R < 10\text{ M}\Omega$, all $C < 100\text{ pF}$.

4: Inverted Zero

The compensator transfer function includes a pole at the origin, which introduces a constant phase shift of -90° . To maintain stability around the LC resonance frequency f_{LC} .

the zero f_{ZEA} should be placed at least one decade below f_{LC} . In this way, the zero starts providing phase boost before the loop reaches the LC double-pole region, preventing the total phase from approaching -180° . In this design, f_{ZEA} is chosen as 2 kHz, which is one decade above the required 200 Hz low-frequency specification. This still ensures sufficient low-frequency gain, such that the loop gain remains greater than 40 dB at 200 Hz. The mid band gain after f_{ZEA} is :

$$A_{mid} = \frac{R_2}{R_1} \cdot PWM_{gain}$$

4: Inverted Zero

By choosing $R_2 = 5R_1$ and $PWM_{\text{gain}} = 2.5$, the midband gain becomes

$$A_{\text{mid}} \approx 22 \text{ dB}$$

and the loop gain at 200 Hz is

$$A_{\text{loop}}(200 \text{ Hz}) \approx 42 \text{ dB}$$

With $C_2 = 90 \text{ pF}$, the required resistor value is

$$R_2 = \frac{1}{2\pi f_{\text{ZEA}} C_2} = 884.1 \text{ k}\Omega \quad (1)$$

4: LHP Zero

The LC filter introduces a second-order pole, which results in a slope of -40 dB/dec and can contribute up to -180° phase shift. To compensate for this phase loss and improve the phase margin, an LHP zero is introduced. This also changes the slope to -20 dB/dec . With $f_{LC} = 50.3\text{ kHz}$, the zero is chosen as

$$f_{FZ} = 45\text{ kHz}$$

The required capacitor value is then

$$C_3 = \frac{5}{2\pi \cdot 45 \times 10^3 \cdot 884.2 \times 10^3} \approx 20\text{ pF} \quad (2)$$

4: Poles f_{FP} and f_{HP}

The poles are placed above the unity gain frequency to limit their phase loss at crossover. Since

$$f_u < \frac{f_{PWM}}{\pi} = 636.6 \text{ kHz},$$

the pole locations are chosen sufficiently high to maintain a phase margin above 60° :

$$f_{FP} = 2 \text{ MHz}, \quad f_{HP} = 4.5 \text{ MHz}.$$

PWM_{gain}	2.5
C_1	40 fF
C_2	90 pF
C_3	20 pF
R_1	$R_2/5$
R_2	884.2 k Ω
R_3	4 k Ω

4: Stability Analysis

The figure below confirms that all specifications are met: a phase margin of 61.2° , a loop gain of 41.1 dB at 200 Hz, and a unity gain frequency of 608.2 kHz.

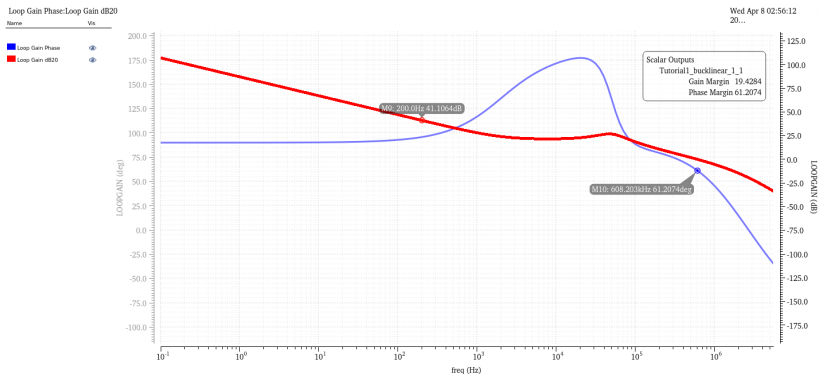


Figure: Loop gain frequency response

5: Transient Stability Verification - Waveforms

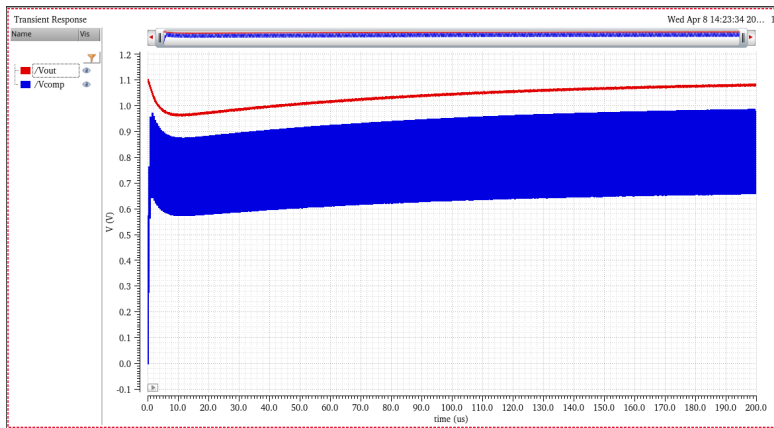


Figure: Transient waveform of V_{out} and V_{ctrl}

5: Transient Stability Verification - Observations

- **Simulation Results:** The output voltage V_{out} and compensator output V_{ctrl} exhibit a well-damped startup response.
- **Observation:** No sustained oscillations or instability are observed after the initial transient. The system settles to the steady-state 1.1V accurately. (*Since the feedback network was configured with $R_{top} = 1k\Omega$ and $R_{bottom} = 10k\Omega$ with a 1V reference, resulting in a target output of 1.1V.*)

6: Input Disturbance Rejection (PSRR)

- **Setup:** $1V_{pp}$ sine wave at 200 Hz added to V_{in} (5 V DC).
- **Observation:** Measured V_{out} ripple is 5.95 mV_{pp} at 200 Hz.
- **Analysis:** Attenuation matches the high loop gain designed in Task 3:

$$\text{PSRR} = 20 \cdot \log_{10} \left(\frac{5.95\text{ mV}}{1\text{ V}} \right) \approx -44.5\text{ dB}$$

- **Conclusion:** Exceeds the $> 40\text{ dB}$ requirement, verifying the low-frequency disturbance rejection.

